EXERCISE

Investment evaluation and deal structuring with a growth equity fund

A growth equity fund is evaluating an investment in a fast-growing digital healthcare platform specialized in AI-based diagnostic solutions.

The investment takes place at time 0, and the fund plans to exit the investment at the end of year 6 (t = 6) through the sale of its stake.

Operating data

- The company is expected to reach at time 6 an EBITDA equal to:
  - EBITDA (time 6) = €25 million
- The firm value is estimated using a fundamentals-based approach, assuming:
  - operating cash flow to EBITDA ratio:
    - $k = 65\%$
  - long-term growth rate:
    - $g = 4\%$
  - discount rate:
    - $r = 16\%$

Investment data

- Initial investment at time 0:
  - €35 million
- During the investment period, the fund receives the following cash flows (dividends):
  - $T_1 = €2\text{million}$
  - $T_2 = €3\text{million}$
  - $T_3 = €4\text{million}$
  - $T_4 = €5\text{million}$
  - $T_5 = €6\text{million}$
  - $T_6 = €7\text{million}$ (before exit)
- Required return (target IRR):
  - 28% per year

Deal structure assumptions

- The investment is fully equity-financed
- No financial leverage is used
- The firm value at time 6 coincides with equity value (no debt)

Questions

0. Implicit multiple

Determine the implied EV/EBITDA multiple using the fundamentals (k, r, g).

1. Firm value

Calculate the value of the firm at time 6.

2. Minimum exit value

Determine the minimum exit value (value of the stake at time 6) required for the fund to achieve its target IRR of 28%, taking into account the intermediate cash flows received.

3. Required ownership

Determine the equity stake the fund must acquire so that the value of its participation at exit equals the required minimum value.

4. Valuation

Determine:

- post-money valuation
- pre-money valuation

5. Convenience for the entrepreneur

Assume that without the external investment the entrepreneur expects the company to reach only:

$$EBITDA_6 = €11 \text{ million}$$

Using the same valuation assumptions:

- determine the firm value without the fund,
- determine whether accepting the investment is economically convenient for the entrepreneur.